1 Forward order for general Williamson coefficients

We show that the commutator-free method (M2) inherits the classical order of the Runge–Kutta method (M1), for arbitrary real coefficients $(A_i, B_i)_{i=1}^s$ and every order p. Two facts drive the proof. First, the piecewise-linear path through the rows of $\mathbf{A}$ and then $\mathbf{b}^\top$, with one segment for each exponential, has a signature whose coefficients are exactly the order conditions of the commutator-free method. Writing the order conditions as signature coefficients lets us use known results about signatures, instead of working in the MKW algebra directly. Second, for Williamson coefficients the signed area between two linear projections of this path is again a linear projection. Since iterated areas generate the shuffle algebra [2], every coefficient of the lifted method is a universal polynomial in the classical elementary weights. The exact flow has the same structure, so the lifted coefficients agree with those of the exact flow whenever the elementary weights do.

1.1 The coefficient path

Let $\boldsymbol{\beta}_i = (\beta_{i1}, \dots, \beta_{is})$ be the ith row of the coefficients (M3), with $\beta_{ij} = 0$ for $j > i$. Equivalently, $\boldsymbol{\beta}_i = B_i \mathbf{w}_i$, where

$$\mathbf{w}_0 = 0, \qquad \mathbf{w}_i = A_i \mathbf{w}_{i-1} + \mathbf{e}_i^\top, \tag{1}$$

and $\mathbf{e}_i^\top$ is the ith coordinate row. By (M4), the rows $\mathbf{a}_i$ of the Butcher matrix $\mathbf{A} = (a_{ij})$ and the weights $\mathbf{b}$ are partial sums of these rows. Setting $\mathbf{a}_{s+1} = \mathbf{b}^\top$, we have

$$\mathbf{a}_i = \sum_{l<i} \boldsymbol{\beta}_l, \qquad 1 \le i \le s+1. \tag{2}$$

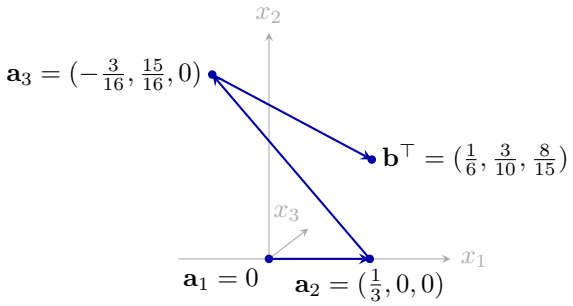
The *coefficient path* $x : [0, s] \to \mathbb{R}^{1\times s}$ is the piecewise-linear path with vertices $x(l) = \mathbf{a}_{l+1}$, $0 \le l \le s$. Its lth segment has increment $\boldsymbol{\beta}_l$. For $u \in \mathbb{R}^s$, write $x^u = xu$ for the scalar projection, and for $u_1, \dots, u_m \in \mathbb{R}^s$ write

$$S_t(u_1 \cdots u_m) = \int_{0<t_1<\cdots<t_m<t} \mathrm{d}x_{t_1}^{u_1} \cdots \mathrm{d}x_{t_m}^{u_m}, \qquad S_t(\emptyset) = 1,$$

for the signature of x paired with $u_1 \otimes \cdots \otimes u_m$. Along the lth segment each x^{u_r} grows linearly, by $\boldsymbol{\beta}_l u_r$, and the iterated integral over that segment alone is $\frac{1}{m!}\prod_r \boldsymbol{\beta}_l u_r$. Integrals over several segments are built from these by Chen's identity, which sorts the ordered times $t_1 < \cdots < t_m$ into the segments they fall in.

Example 1.1. Williamson's three-stage scheme of classical order three has coefficients $A_i = 0, -\frac{5}{9}, -\frac{153}{128}$ and $B_i = \frac{1}{3}, \frac{15}{16}, \frac{8}{15}$, so $\boldsymbol{\beta}_1 = (\frac{1}{3}, 0, 0)$, $\boldsymbol{\beta}_2 = (-\frac{25}{48}, \frac{15}{16}, 0)$ and $\boldsymbol{\beta}_3 = (\frac{17}{48}, -\frac{51}{80}, \frac{8}{15})$. The tableau and coefficient path are

0	0	0	0
$\frac{1}{3}$	$\frac{1}{3}$	0	0
$\frac{3}{4}$	$-\frac{3}{16}$	$\frac{15}{16}$	0
	$\frac{1}{6}$	$\frac{3}{10}$	$\frac{8}{15}$



Recall that $g_{l+1}(B_+(\tau_1 \cdots \tau_k))$ is the Lie–Butcher coefficient of the forest $\tau_1 \cdots \tau_k$ in the stage Y_l, reached after l exponentials. For $l = s$ it is $\phi(\tau_1 \cdots \tau_k)$, the coefficient in the MKW order condition for that forest. For a tree τ, let $\mathbf{g}(\tau) = (g_1(\tau), \dots, g_s(\tau))^\top$ be its coefficients in the s slopes.

By (M5), the lth exponential contains τ with coefficient $\mathcal{V}_l(\tau) = \boldsymbol{\beta}_l \mathbf{g}(\tau)$, which is how much $x^{\mathbf{g}(\tau)}$ grows on the lth segment. Each exponential therefore acts as one straight segment of the path, and composing exponentials corresponds to appending segments.

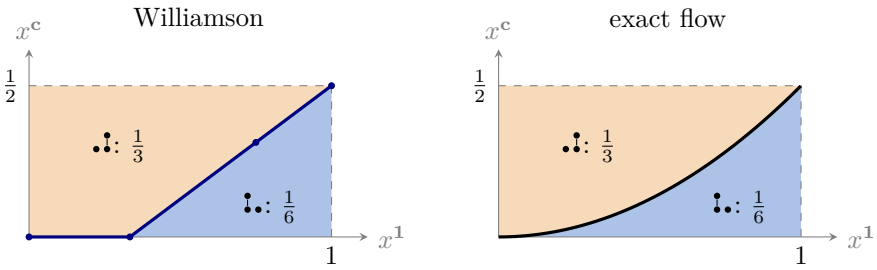
Lemma 1.2 (Lie–Butcher coefficients as signatures)**.** *For $0 \le l \le s$ and $\tau_1, \dots, \tau_k \in \mathcal{T}^{\mathrm{pl}}$,*

$$g_{l+1}(B_+(\tau_1 \cdots \tau_k)) = S_l\big(\mathbf{g}(\tau_1) \cdots \mathbf{g}(\tau_k)\big).$$

In particular, $\phi(\tau_1 \cdots \tau_k) = S_s(\mathbf{g}(\tau_1) \cdots \mathbf{g}(\tau_k))$.

Proof. The increment of $x^{\mathbf{g}(\tau)}$ over the lth segment is $\boldsymbol{\beta}_l \mathbf{g}(\tau) = \mathcal{V}_l(\tau)$. Hence (M5) is the signature of the lth segment, and (M6) is Chen's identity for appending it. Both sides equal $\varepsilon(\tau_1 \cdots \tau_k)$ when $l = 0$. □

Example 1.3. In Example 1.1, the nodes are $\mathbf{c} = (0, \frac{1}{3}, \frac{3}{4})$. By Lemma 1.2, the order conditions for ⁝• and •⁝ are $S_3(\mathbf{c}\,\mathbf{1}) = \int x^{\mathbf{c}}\, \mathrm{d}x^{\mathbf{1}} = \frac{1}{6}$ and $S_3(\mathbf{1}\,\mathbf{c}) = \int x^{\mathbf{1}}\, \mathrm{d}x^{\mathbf{c}} = \frac{1}{3}$: the areas below and to the left of the path in the $(x^{\mathbf{1}}, x^{\mathbf{c}})$-plane.



The paths differ, but the areas agree, as Theorem 1.8 predicts.

The simplest cases show what the lemma produces. With a single tree the integral is just the total growth of the path, so $\mathbf{g}(B_+(\tau)) = \mathbf{A}\mathbf{g}(\tau)$; for $\tau = \bullet$ this gives the vector of nodes, $\mathbf{g}(⁝) = \mathbf{c} = \mathbf{A}\mathbf{1}$. With the same tree repeated, the integral is a power divided by a factorial; for example, $\mathbf{g}(\curlyvee) = \frac{1}{2}\mathbf{c} \odot \mathbf{c}$. The order of the exponentials matters only when different trees appear, since in general $S_l(\mathbf{c}\,\mathbf{1}) \neq S_l(\mathbf{1}\,\mathbf{c})$. Lemma 1.4 below is what controls these terms.

1.2 Areas are path coordinates

Let $\mathcal{T}$ be the set of (non-planar) rooted trees, with $\tau! = |\tau|\,\tau_1! \cdots \tau_k!$ for $\tau = B_+(\tau_1 \cdots \tau_k)$ and $\bullet! = 1$. The elementary weight vectors $\boldsymbol{\Phi}(\tau) \in \mathbb{R}^s$ are

$$\boldsymbol{\Phi}(\bullet) = \mathbf{1}, \qquad \boldsymbol{\Phi}(B_+(\tau_1 \cdots \tau_k)) = \mathbf{A}\boldsymbol{\Phi}(\tau_1) \odot \cdots \odot \mathbf{A}\boldsymbol{\Phi}(\tau_k),$$

where $\odot$ is the componentwise product, and (M1) has classical order p if and only if $\mathbf{b}^\top \boldsymbol{\Phi}(\tau) = 1/\tau!$ for all $\tau \in \mathcal{T}$ with $|\tau| \le p$. For $u, v \in \mathbb{R}^s$ define

$$u \triangleright v = (\mathbf{A}u) \odot v, \qquad D(u, v) = u \triangleright v - v \triangleright u.$$

If $\sigma \curvearrowright \tau = B_+(\tau_1 \cdots \tau_k \sigma)$ denotes grafting σ onto the root of $\tau = B_+(\tau_1 \cdots \tau_k)$, the definition of $\boldsymbol{\Phi}$ gives $\boldsymbol{\Phi}(\sigma) \triangleright \boldsymbol{\Phi}(\tau) = \boldsymbol{\Phi}(\sigma \curvearrowright \tau)$. Extending $\boldsymbol{\Phi}$ linearly,

$$D(\boldsymbol{\Phi}(\sigma), \boldsymbol{\Phi}(\tau)) = \boldsymbol{\Phi}(\sigma \curvearrowright \tau - \tau \curvearrowright \sigma). \tag{3}$$

For scalar paths f, g with $f(0) = g(0) = 0$, write $\mathrm{Area}(f, g)_t = \int_0^t f\, \mathrm{d}g - g\, \mathrm{d}f$, twice the Lévy area of the planar path (f, g).

Lemma 1.4 (Areas are path coordinates)**.** *For all $u, v \in \mathbb{R}^s$, $\mathrm{Area}(x^u, x^v) = x^{D(u,v)}$ on $[0, s]$.*

Proof. On the ith segment, $x = \mathbf{a}_i + \theta\boldsymbol{\beta}_i$ with $\theta \in [0, 1]$, so the terms in θ cancel in the signed-area contribution

$$x^u \, \mathrm{d}x^v - x^v \, \mathrm{d}x^u = \big((\mathbf{a}_i u)(\boldsymbol{\beta}_i v) - (\mathbf{a}_i v)(\boldsymbol{\beta}_i u)\big)\mathrm{d}\theta.$$

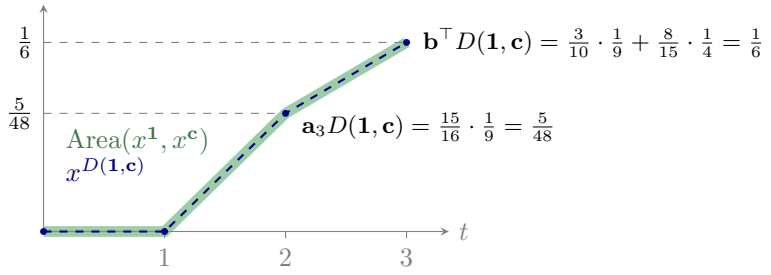
Since $\mathrm{d}x^{D(u,v)} = \boldsymbol{\beta}_i D(u, v)\,\mathrm{d}\theta$ and both sides of the lemma vanish at $t = 0$, it suffices to prove

$$r\, D(u, v) = (\mathbf{a}_i u)(rv) - (\mathbf{a}_i v)(ru) \tag{4}$$

for $r = \boldsymbol{\beta}_i$, or equivalently for $r = \mathbf{w}_i$. Both sides are linear in r, and $r = \mathbf{e}_i^\top$ satisfies (4) because $(\mathbf{A}u)_i = \mathbf{a}_i u$; this settles $i = 1$. For $i \geq 2$, assume (4) for $r = \mathbf{w}_{i-1}$ with $\mathbf{a}_{i-1}$ in place of $\mathbf{a}_i$. Replacing $\mathbf{a}_{i-1}$ by $\mathbf{a}_i = \mathbf{a}_{i-1} + B_{i-1}\mathbf{w}_{i-1}$ adds $B_{i-1}\big((\mathbf{w}_{i-1}u)(\mathbf{w}_{i-1}v) - (\mathbf{w}_{i-1}v)(\mathbf{w}_{i-1}u)\big) = 0$, so $r = \mathbf{w}_{i-1}$ also satisfies (4). Linearity and (1) give it for $\mathbf{w}_i = A_i\mathbf{w}_{i-1} + \mathbf{e}_i^\top$. □

By (3), Lemma 1.4 says that the area of two tree projections of x is the projection of the commutator of grafting. Because the identity holds on all of $[0, s]$, it can be iterated.

Example 1.5. In Example 1.1, $D(\mathbf{1}, \mathbf{c}) = \mathbf{c} \odot \mathbf{c} - \mathbf{A}\mathbf{c} = (0, \frac{1}{9}, \frac{1}{4})^\top$. By Lemma 1.4, the accumulated area $\mathrm{Area}(x^{\mathbf{1}}, x^{\mathbf{c}})$ equals the path coordinate $x^{D(\mathbf{1},\mathbf{c})}$ at every time:



At $t = 3$ both equal $\mathbf{b}^\top(\mathbf{c} \odot \mathbf{c}) - \mathbf{b}^\top\mathbf{A}\mathbf{c} = \frac{1}{3} - \frac{1}{6}$, a combination of classical elementary weights.

1.3 Reduction to elementary weights

Call the scalar paths $z^1, \ldots, z^n$ *iterated areas of order one*, and $\mathrm{Area}(f, g)$ an iterated area whenever f and g are. We use the following consequence of the result of Diehl, Lyons, Preiß and Reizenstein.

Theorem 1.6 ([2, Corollary 4.8])**.** *For every word $j_1 \cdots j_m$ in $\{1, \ldots, n\}$ there is a polynomial P with rational coefficients, in the values at T of finitely many iterated areas regarded as formal bracket expressions, such that*

$$\int_{0<t_1<\cdots<t_m<T} \mathrm{d}z^{j_1}_{t_1} \cdots \mathrm{d}z^{j_m}_{t_m} = P$$

for all piecewise polynomial paths $z^1, \ldots, z^n : [0, T] \to \mathbb{R}$ starting at zero. Each monomial of P contains each letter j exactly $\#\{r : j_r = j\}$ times.

Diehl et al. show that iterated areas generate the shuffle algebra. Since the signature is a shuffle character that maps the area operation on words to Area, this gives Theorem 1.6. The area operation preserves the number of occurrences of each letter, so P may be replaced by its homogeneous component.

Let M_σ, $\sigma \in \mathcal{T}$, be indeterminates of degree $|\sigma|$. For a polynomial Q in these indeterminates and a row $r \in \mathbb{R}^{1\times s}$, write $Q[r]$ for its value at $M_\sigma = r\boldsymbol{\Phi}(\sigma)$. In particular, $Q[\mathbf{b}^\top]$ is a polynomial in the elementary weights.

Proposition 1.7. *For $\tau_1, \dots, \tau_m \in \mathcal{T}$ there is a polynomial $Q_{\tau_1\cdots\tau_m}$, homogeneous of degree $|\tau_1| + \cdots + |\tau_m|$ and independent of s and (A_i, B_i), such that*

$$S_t\big(\boldsymbol{\Phi}(\tau_1)\cdots\boldsymbol{\Phi}(\tau_m)\big) = Q_{\tau_1\cdots\tau_m}[x(t)], \qquad 0 \le t \le s.$$

Proof. Apply Theorem 1.6 to $z^j = x^{\boldsymbol{\Phi}(\tau_j)}$ on $[0,t]$. By Lemma 1.4 and (3), each iterated area is $x^{\boldsymbol{\Phi}(\eta)}$, where η is a linear combination of trees obtained by iterated grafting commutators of the τ_j it contains. Its value at t is therefore a linear combination of $x(t)\boldsymbol{\Phi}(\sigma)$ with $|\sigma|$ equal to the total size of those τ_j. Substituting into P gives $Q_{\tau_1\cdots\tau_m}$, which depends only on P and the grafting commutators. □

We extend $Q_{\tau_1\cdots\tau_m}$ multilinearly to linear combinations of trees.

The proof uses only Theorem 1.6, Lemma 1.4 and the definition of $\boldsymbol{\Phi}$. It therefore applies to any linear map $\mathbf{A}$ on a commutative unital algebra, together with a piecewise polynomial path x of linear functionals starting at zero, that satisfy Lemma 1.4. The exact flow is of this form. Let $\mathbf{A}$ act on polynomials by $(\mathbf{A}u)(\rho) = \int_0^\rho u$, let $\odot$ be the ordinary product, and let $x(t)u = \int_0^t u$ for $t \in [0,1]$. Then

$$\boldsymbol{\Phi}(\tau)(\rho) = \frac{|\tau|\,\rho^{|\tau|-1}}{\tau!}, \qquad x(t)\boldsymbol{\Phi}(\tau) = \frac{t^{|\tau|}}{\tau!},$$

by induction on τ. Lemma 1.4 holds here because both sides have derivative $(\mathbf{A}u)v - (\mathbf{A}v)u$. Write $Q[\mathrm{ex}_t]$ for the value of Q at $M_\sigma = x(t)\boldsymbol{\Phi}(\sigma) = t^{|\sigma|}/\sigma!$. The proof of Proposition 1.7 then gives, with the same polynomials and for linear combinations η_j of trees,

$$Q_{\eta_1\cdots\eta_m}[\mathrm{ex}_t] = \int_{0<t_1<\cdots<t_m<t} \boldsymbol{\Phi}(\eta_1)(t_1)\cdots\boldsymbol{\Phi}(\eta_m)(t_m)\,\mathrm{d}t_1\cdots\mathrm{d}t_m. \tag{5}$$

1.4 Order transfer

The exact flow has coefficients $\phi_{\mathrm{ex}}(\omega)$, $\omega \in \mathcal{F}^{\mathrm{pl}}$, given by $\phi_{\mathrm{ex}}(\emptyset) = 1$ and

$$\phi_{\mathrm{ex}}(\tau_1\cdots\tau_k) = \frac{\phi_{\mathrm{ex}}(\omega_1)\cdots\phi_{\mathrm{ex}}(\omega_k)}{\prod_{j=1}^k(|\tau_1| + \cdots + |\tau_j|)}, \qquad \tau_j = B_+(\omega_j), \tag{6}$$

in the normalisation of Proposition M10 [3, Corollary 2.6]. For example, $\phi_{\mathrm{ex}}(\bullet) = 1$ and $\phi_{\mathrm{ex}}(\begin{smallmatrix}\bullet\\\bullet\end{smallmatrix}) = \phi_{\mathrm{ex}}(\bullet\!\!-\!\!\bullet) = \frac{1}{2}$, recovering (M11).

For a monomial $\mu = M_{\sigma_1}\cdots M_{\sigma_r}$ write $B_+(\mu) = B_+(\sigma_1\cdots\sigma_r) \in \mathcal{T}$, and extend B_+ linearly to polynomials. Since $(\mathbf{A}\boldsymbol{\Phi}(\sigma))_i = \mathbf{a}_i\boldsymbol{\Phi}(\sigma)$,

$$\big(Q[\mathbf{a}_i]\big)_{i=1}^s = \boldsymbol{\Phi}(B_+(Q)). \tag{7}$$

Theorem 1.8 (Order transfer)**.** *Let $p \ge 1$. If the Runge–Kutta method (M1) has classical order p, then $\phi(\omega) = \phi_{\mathrm{ex}}(\omega)$ for every $\omega \in \mathcal{F}^{\mathrm{pl}}$ with $|\omega| \le p$. Consequently, if $\xi \in C^{p+1}$, the commutator-free method (M2) has forward order p.*

Proof. We show by induction on $|\tau|$ that every $\tau = B_+(\omega) \in \mathcal{T}^{\mathrm{pl}}$ has a polynomial Q_τ, homogeneous of degree $|\omega|$ and independent of s and (A_i, B_i), such that

$$g_i(\tau) = Q_\tau[\mathbf{a}_i] \quad (1 \le i \le s+1), \qquad Q_\tau[\mathrm{ex}_t] = t^{|\omega|}\phi_{\mathrm{ex}}(\omega).$$

For $\tau = \bullet$ take $Q_\tau = 1$. Let $\tau = B_+(\tau_1 \cdots \tau_k)$, assume the claim for each τ_j, and set $\eta_j = B_+(Q_{\tau_j})$, a linear combination of trees with $|\tau_j|$ vertices. By (7), $\mathbf{g}(\tau_j) = \mathbf{\Phi}(\eta_j)$. Lemma 1.2 and Proposition 1.7, expanded multilinearly, give

$$g_{l+1}(\tau) = S_l\big(\mathbf{\Phi}(\eta_1)\cdots\mathbf{\Phi}(\eta_k)\big) = Q_{\eta_1\cdots\eta_k}[x(l)] = Q_{\eta_1\cdots\eta_k}[\mathbf{a}_{l+1}],$$

so $Q_\tau = Q_{\eta_1\cdots\eta_k}$ has the stated degree $|\tau_1|+\cdots+|\tau_k| = |\omega|$. For the exact flow, $\mathbf{\Phi}(B_+(\mu))(\rho) = \mu[\mathrm{ex}_\rho]$ for each monomial μ, so the inductive hypothesis gives $\mathbf{\Phi}(\eta_j)(\rho) = \rho^{|\tau_j|-1}\phi_{\mathrm{ex}}(\omega_j)$. Then (5) and (6) give

$$Q_\tau[\mathrm{ex}_t] = \int_{0<t_1<\cdots<t_k<t} \prod_{j=1}^{k} \phi_{\mathrm{ex}}(\omega_j)\, t_j^{|\tau_j|-1}\,\mathrm{d}t_1\cdots\mathrm{d}t_k = t^{|\omega|}\phi_{\mathrm{ex}}(\omega).$$

Now let $|\omega| \le p$. Every M_σ in $Q_{B_+(\omega)}$ has $|\sigma| \le |\omega| \le p$, so the classical order conditions give $Q_{B_+(\omega)}[\mathbf{b}^\top] = Q_{B_+(\omega)}[\mathrm{ex}_1]$. By (M7), $\phi(\omega) = g_{s+1}(B_+(\omega)) = \phi_{\mathrm{ex}}(\omega)$. The error bound follows from Proposition M10 as in the proof of Theorem M9. □

In particular, Theorem M8 and Theorem 1.8 give forward order two for CF-EES$_{\mathrm{c}}(2, 2s-1)$ for every $s \ge 2$.

Example 1.9. For CF-EES$(2,5)$, the coefficients (M12) give $\boldsymbol{\beta}_1 = (\frac{1}{2}, 0, 0)$, $\boldsymbol{\beta}_2 = (-\frac{1}{2}, 1, 0)$, $\boldsymbol{\beta}_3 = (\frac{1}{4}, -\frac{1}{2}, \frac{1}{4})$ and $\mathbf{c} = (0, \frac{1}{2}, 1)$. Of the coefficients below, only $\phi(\begin{smallmatrix}\bullet\\ \bullet\bullet\end{smallmatrix}) = S_3(\mathbf{c}\,\mathbf{1})$ depends on the order of the exponentials. By the shuffle identity and Lemma 1.4,

$$2S_3(\mathbf{c}\,\mathbf{1}) = (\mathbf{b}^\top\mathbf{c})(\mathbf{b}^\top\mathbf{1}) + \mathbf{b}^\top D(\mathbf{c}, \mathbf{1}).$$

The polynomials $Q_{B_+(\omega)}$ and their values for $|\omega| \le 3$ are

ω	$Q_{B_+(\omega)}$	$\phi(\omega)$	$\phi_{\mathrm{ex}}(\omega)$
$\bullet$	$M_{\bullet}$	1	1
$\begin{smallmatrix}\bullet\\ \bullet\end{smallmatrix}$	$M_{\begin{smallmatrix}\bullet\\ \bullet\end{smallmatrix}}$	$\frac{1}{2}$	$\frac{1}{2}$
$\bullet\bullet$	$\frac{1}{2}M_{\bullet}^2$	$\frac{1}{2}$	$\frac{1}{2}$
$\begin{smallmatrix}\bullet\\ \bullet\\ \bullet\end{smallmatrix}$	$M_{\begin{smallmatrix}\bullet\\ \bullet\\ \bullet\end{smallmatrix}}$	$\frac{1}{8}$	$\frac{1}{6}$
$\begin{smallmatrix} & \bullet & \\ \bullet & & \bullet\end{smallmatrix}$	$\frac{1}{2}M_{\begin{smallmatrix} & \bullet & \\ \bullet & & \bullet\end{smallmatrix}}$	$\frac{3}{16}$	$\frac{1}{6}$
$\begin{smallmatrix}\bullet\\ \bullet\bullet\end{smallmatrix}$	$\frac{1}{2}\big(M_{\bullet}M_{\begin{smallmatrix}\bullet\\ \bullet\end{smallmatrix}} + M_{\begin{smallmatrix}\bullet\\ \bullet\\ \bullet\end{smallmatrix}} - M_{\begin{smallmatrix} & \bullet & \\ \bullet & & \bullet\end{smallmatrix}}\big)$	$\frac{1}{8}$	$\frac{1}{6}$

The first three rows are the forests of degree at most two, so the lift has forward order two. The remaining rows fail at degree three, and the values of ϕ agree with Table M13.

References

[1] A. Bazavov, *Commutator-free Lie group methods with minimum storage requirements and reuse of exponentials*, BIT Numer. Math. 62 (2022), 745–771.

[2] J. Diehl, T. Lyons, R. Preiß, and J. Reizenstein, *Areas of areas generate the shuffle algebra*, arXiv:2002.02338 (2021).

[3] B. Owren, *Order conditions for commutator-free Lie group methods*, J. Phys. A 39 (2006), 5585–5599.